

B.Tech. Degree II Semester Regular Examination July 2024**EE/EC/CS/IT 23-200-0202B ENGINEERING CHEMISTRY**

(2023 Scheme)

Time: 3 Hours

Maximum Marks: 50

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Explain the basic concepts of chemical thermodynamics and quantum chemistry.
 CO2: Illustrate the spectroscopic methods in characterizing materials.
 CO3: Develop electro chemical methods to protect different metals from corrosion.
 CO4: Interpret the chemistry of a few important engineering materials and their industrial applications.
 CO5: Understand the principle, concept, working and applications of relevant technologies and comparison of results with theoretical calculations.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A(Answer *ALL* questions)

	(5 × 2 = 10)	Marks	BL	CO	PO
I. (a)	Discuss the thermodynamics of biochemical reactions.	2	L1	1	3
(b)	An electron is confined to move in a one-dimensional box of length 1nm undergoes transition from the first excited state to the second excited state. Calculate the transition energy in eV.	2	L3	5	2
(c)	List out the possible electronic transitions in the following molecules: (i) CH ₃ SH (ii) CH ₂ =CH ₂ (iii) Phenol (iv) Cyclobutane.	2	L2	2	3
(d)	Write a note on viscosity index of lubricants.	2	L1	4	2
(e)	Explain the prevention methods for corrosion.	2	L1	3	2

PART B

(4 × 10 = 40)

- II. (a) Derive the expression connecting free energy (ΔG) and equilibrium constant. 7 L2 1 3
- (b) Carbon monoxide, a toxic product from the incomplete combustion of fossil fuels, reacts with water to form CO₂ and H₂, as shown in the equation $\text{CO(g)} + \text{H}_2\text{O(g)} \rightleftharpoons \text{CO}_2\text{(g)} + \text{H}_2\text{(g)}$, for which $\Delta H^\circ = -41.0 \text{ kJ/mol}$ and $\Delta S^\circ = -42.3 \text{ J (mol} \cdot \text{K)}$ at 25°C and 1 atm.
- (i) What is ΔG° for this reaction?
- (ii) What is ΔG if the gases have the following partial pressures:
 $P_{\text{CO}} = 1.3 \text{ atm}$, $P_{\text{H}_2\text{O}} = 0.8 \text{ atm}$, $P_{\text{CO}_2} = 2.0 \text{ atm}$, and $P_{\text{H}_2} = 1.3 \text{ atm}$? 3 L3 1 3

OR

- III. Explain the phase diagram of water with a neat labelled diagram. 10 L2 1 3

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		Marks	BL	CO	PO
IV.	Write the Schrodinger wave equation for hydrogen atom in terms of polar coordinates. Separate the resultant equation in three equations using the technique of separation of variables. How do the quantum numbers n , l and m emerge from the solution of wave equation?	10	L2	5	3
OR					
V.	Derive the energy expression for a particle of mass 'm' confined to move in a one dimensional box of length 'a'. Find out the corresponding wave function.	10	L2	5	3
VI. (a)	Draw the labelled low and high resolution ^1H NMR spectra for (i) $\text{CF}_2=\text{CHF}$ (ii) $\text{CH}_2-(\text{CH}_2\text{SH})_2$ (iii) $\text{CH}_3-\text{CH}(\text{Cl})-\text{CH}_3$.	6	L3	2	2
(b)	The pure rotational spectrum of gaseous molecule CN consists of a series of equally spaced lines separated by 3.798 cm^{-1} . Calculate the internuclear distance of the molecule. The molar masses are $^{12}\text{C} = 12.011\text{ g mol}^{-1}$ and $^{14}\text{N} = 14.007\text{ g mol}^{-1}$.	4	L3	2	2
OR					
VII. (a)	Which type of nuclei are NMR active? Under what conditions such nuclei will be in resonance with radiation absorbed?	6	L3	2	2
(b)	Illustrate with the help of a diagram IR active vibrations of CO_2 and H_2O molecule.	4	L3	2	2
VIII. (a)	Write a note Fullerene and Graphene.	5	L1	4	3
(b)	Derive the expression connecting electrode potential and metal ion concentration.	5	L1	3	3
OR					
IX. (a)	Write a note on conducting polymers.	5	L1	4	3
(b)	Discuss the working principle of lead storage battery.	5	L1	3	3

Bloom's Taxonomy Levels

L1 – 38.8%, L2 – 27.8%, L3 – 33.4%.
